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ReefSafe decision dashboard

Observed reef evidence + cautious next-observation screening | Malaysia, 2012–2025

Selected island

Labuan

Change the yellow cell to inspect any monitored island

Current LCC

24.2%

Observed change

-7.3 pp/yr

Predicted next change

-12.0 pp/yr

Priority

High screening priority

Uncertainty interval

-19.8 to -4.2 pp/yr

Intervals are empirical screening bands, not causal or site-specific certainty.

Observed evidence

NOAA thermal stress observed

Recommended next step

Coordinate bleaching survey; do not attribute thermal loss to visitors

Top 10 screening priorities: predicted next change (pp/yr)

Labuan

Kapas

Lankayan

Seri Buat

Mertang

Mensirip

Tenggol

Sibu

Tinggi

Tun Sakaran Marine Park

Observed national mean live coral cover (%)

2012

2013

2014

2015

2016

2017

2018

2019

2020

2021

2022

2023

2024

2025

Island	State	Current LCC	Predicted change	Evidence
Labuan	Sabah	24.2	-12.0	NOAA thermal stress observed
Kapas	Terengganu	52.6	-4.5	Modelled decline; no single dominant observed stressor
Lankayan	Sabah	51.0	-3.8	NOAA thermal stress observed
Seri Buat	Pahang	54.2	-3.6	Anchor impact mentioned in Reef Check report
Mertang	Johor	58.3	-3.4	Anchor impact mentioned in Reef Check report
Mensirip	Johor	54.2	-3.2	Waste or pollution indicator is elevated
Tenggol	Terengganu	54.2	-3.1	Waste or pollution indicator is elevated
Sibu	Johor	52.8	-3.1	Anchor impact mentioned in Reef Check report
Tinggi	Johor	52.5	-3.0	Anchor impact mentioned in Reef Check report
Tun Sakaran Marine Park	Sabah	51.8	-3.0	NOAA thermal stress observed

Model	MAE	RMSE	R²
Baseline mean	5.991	8.100	-0.019
Ridge regression	5.993	7.909	0.029
Gradient boosting	5.902	7.817	0.051
Random forest	6.123	8.073	-0.012

Model check

Gradient boosting MAE 5.90 pp/yr vs baseline 5.99 pp/yr (+1.5%). R² 0.051. Use only to rank field checks; never as proof of tourism causality.